MANAS AGGRAWAL

Boston, MA | (857) 265-1533 | aggrawal.m@northeastern.edu | Portfolio | Linkedin | Github | Medium | Leetcode

EDUCATION

Northeastern University — **Boston, MA**
Masters of Science – Computer Science | GPA: 3.86 — *Sep 2024 – Dec 2026*

- Graduate Teaching Assistant for Fundamentals of Software Engineering (CS 4530)
- Researched Multi-Agent LLM architectures, for e2e SDLC automation, across accuracy, cost efficiency

WORK EXPERIENCE

Hydrow — **Boston, MA**
Software Engineer Intern — *Jan 2026 – Aug 2026*

- Kept every user-facing endpoint under 25 ms at production scale for several product features end-to-end — schema, APIs, tests, and rollout — including the strength-fitness app's progress graphs
- Cut a real-time badge eligibility check from ~2,000 ms to near-zero and an event-loop-blocking graph computation to 0.7 ms by pushing data-heavy work into SQL with materialized views and Redis caching
- Restored data correctness across ~123K historical records via a bounded, dry-run-verified backfill migration, and kept large-table reads scale-safe with cursor-based pagination
- Built an AI-assisted development workflow on Claude Code that automates the coding, testing, and self-review stages of the SDLC while keeping me in the loop for planning and final review, backed by a persistent knowledge base of accumulated review standards
- Eliminated dead-end workout recommendations across all equipment and difficulty combinations by building a multi-pass SQL pipeline that relaxes constraints stepwise until it returns a viable workout
- Reduced new-badge rollout from a coordinated frontend + backend change to backend-only across 21 badge types by shipping a schema-driven API that generates admin forms from validation schemas, plus 5+ real-time badge checkers
- Enabled per-segment content delivery without redeploys by designing a pluggable A/B framework with a master/leaf tile mechanism targeted on churn score and behavioral segment

Studio Graphene — **Gurgaon, India**
Forward Deployed Engineer — *Nov 2020 – Jul 2024*

- Boosted engineering velocity 23% by building a serverless analytics platform on AWS Lambda that pinpointed PR wait times, build failures, and blocked dependencies
- Served 2M+ daily requests against a 500GB+ catalog by designing a Node.js/TypeScript REST backend with PostgreSQL partitioning/indexing and real-time cross-region inventory sync
- Reduced response latency ~40% by moving ingestion and processing to event-driven microservices on AWS SQS

PROJECTS

Nodejs Observability | ***Nodejs, TypeScript, AWS X-Ray, npm, OpenTelemetry, cloudwatch, Jaeger, Prometheus***

- A vendor-agnostic OpenTelemetry package with pluggable presets for AWS, Honeycomb, Jaeger, Tempo, or any OTLP-native backend. Ships traces, metrics, and logs through a single sidecar collector with cost-mindful defaults. Reduces mean debugging time by ~60%
- It has over 1500 downloads on npm

Huddle: Multiplayer AI Agent Workspace | ***TypeScript, NestJS, React, PostgreSQL, Yjs, Socket.IO, Vercel AI SDK, Docker***

- Shared live workspace where a team and an agent research and co-write one document. Every agent step, tool call, message and decision streams to all members in real time, built on an append-only timeline of 30+ event types that also powers session replay and "since you left" catch-up
- Let teams safely steer an agent mid-run, verified by 201 end-to-end tests across 28 suites, by designing role-based controls (Owner/Editor/Viewer), Suggest/Interrupt/Ask message modes, 5 kinds of human approval gates, and conflict detection that sends contradictory instructions to the Owner instead of letting the agent pick one

TECHNICAL SKILLS

Languages:	JavaScript, TypeScript, Java, C++, Python
Frameworks:	Node.js, Express.js, NestJS, Django, Laravel, React
Database:	SQL, Postgres, MySQL, NoSQL, MongoDB, Elasticsearch, Amazon DynamoDB, Firebase
Cloud & DevOps:	Cloud Computing, AWS (Lambda, S3, SQS, DLQ, X-Ray), Docker, CI/CD, cloud infrastructure
Others:	Git, Sentry, OpenTelemetry, linux, REST APIs, web architecture, distributed systems, agile, SDLC